

The Economics of Bestsellers: Consumer Search, Sales Ranking, and Social Learning

Wentao Lu,^a Man Yu^{b,*}
^aFaculty of Business for Science & Technology, School of Management, University of Science and Technology of China, Anhui 230052, China;

^bSchool of Business and Management, Hong Kong University of Science and Technology, Hong Kong

*Corresponding author

Contact: luwentao@ustc.edu.cn,  <https://orcid.org/0000-0001-9338-0441> (WL); manyu@ust.hk,  <https://orcid.org/0000-0002-5035-8073> (MY)

Received: October 26, 2023

Revised: December 31, 2024; August 10, 2025;
September 20, 2025; December 15, 2025

Accepted: December 16, 2025

Published Online in Articles in Advance:
February 4, 2026

<https://doi.org/10.1287/msom.2023.0583>

Copyright: © 2026 INFORMS

Abstract. *Problem definition:* Motivated by major e-commerce platforms' diverse practices in bestseller information provision, this paper examines consumers' learning, searching, and purchasing behavior under uncertainty about products' values, whereas a revenue-maximizing platform strategically decides whether and, if so, how to disclose products' past sales information to consumers. *Methodology/results:* We analyze a two-period Bayesian learning model that embeds consumers' sequential product search in a social learning framework and shows how the interaction between bestseller information and consumer search impacts sales and welfare. We find that a bestseller list constitutes an informative yet noisy signal about the products' values. The informativeness of the signal is determined by the granularity of the bestseller information. By evaluating bestseller information of two levels of granularity, sales ranking and sales volume, we discover that, although consumers benefit more from information of a higher granularity (i.e., sales volume), the platform may prefer providing information of a lower granularity (i.e., sales ranking), suggesting that the platform may withhold information at the cost of consumers. In particular, an inference effect unique to multiproduct Bayesian learning gives rise to the possibility that disclosure of sales volume backfires and hurts the platform. We demonstrate significant sales implications of bestseller information granularity and show that concave distribution functions for consumers' search cost, a stochastic increase in product values, or a growth in consumer population can tilt the platform's preference toward displaying bestseller rankings without revealing sales volumes. Furthermore, we show that bestseller information may lead to lower purchased value or higher search cost, the latter implying that public learning may stimulate rather than substitute private learning. *Managerial implications:* The paper cautions retail platform practitioners about a pitfall associated with disclosing bestseller sales volume and presents guidelines on the timing and granularity of sales information provision. The findings also suggest e-commerce platforms with consumer-centric goals enhance bestseller information transparency on their marketplaces.

Funding: W. Lu gratefully acknowledges support from Johns Hopkins Carey Business School throughout his postdoctoral appointment. The work of M. Yu was partially supported by the Hong Kong Research Grant Council [Grant 16505020] and the National Natural Science Foundation of China [Grant 72022023].

Supplemental Material: The online supplement is available at https://isom.hkust.edu.hk/sites/isom/files/people/sales_ranking_supplement_final.pdf.

Keywords: sequential search • bestseller • social learning

1. Introduction

With the proliferation of new products in the digital era, navigating myriad goods to choose the right one for purchase is a daunting task to many consumers. That is why many shoppers refer to bestseller lists for assistance. A bestseller list presents the most popular items based on unit sales and exposes to potential buyers (otherwise invisible) past purchases of other consumers. The practice of offering bestseller lists can be dated back to 1895 (Bassett and Walter 2001) with

notable examples including *The New York Times* bestseller list for books and IMDb's top box office chart of movies. Also well received by consumers is Amazon.com's bestseller list, which is updated hourly for a wide range of product categories, including appliances, electronics, software, toys, and games.¹ Bestseller lists as a positive stimulus to sales are well-documented by empirical studies in various markets. Sorensen (2007) shows that appearing on *The New York Times* bestseller list can increase book sales by roughly 13%, on average,

and the number can be as high as 57% for debut authors. For restaurants, Cai et al. (2009) demonstrate that informing diners about the rankings of the five most popular dishes substantially boosts the demand for those dishes. Carare (2012) examines empirical evidences from Apple's App Store and identifies a causal effect of past bestseller ranking on current software sales.

The power of bestseller lists lies in the fact that they create an opportunity for peer observation and, thus, facilitate social learning. Wilson et al. (2015) note that observing other consumers is "very common" (p. 2) and "strikingly important" (p. 3) when consumers decide what to buy, and peer observation is even more important than word of mouth for some products. In particular, for search goods, other consumers' purchasing choices may be indicative of different products' values because, before making their choices, these consumers likely have spent time and efforts to research and compare across the products. Hence, a bestseller list allows a consumer to gain insights about the product knowledge that other consumers have acquired through their product searches and, in turn, impacts the consumer's behavior in the consumer's own search for the best product. This is evidenced by Ursu (2018), who analyzes data from a field experiment at Expedia and quantifies a causal effect of rankings on consumer search.

Whereas bestseller information is widely observed for various products and services, this paper focuses on search products in e-commerce markets. Despite the importance of bestseller information to consumers' search and purchasing decisions, how and when the information is accessible to consumers are usually beyond their control. Instead, they are often at the discretion of shopping platforms. Intriguingly, major shopping platforms do not seem to agree upon how much bestseller information to share with consumers. For example, as shown in the left panel of Figure 1,

Amazon.com presents its bestsellers of each prespecified category (e.g., electronics) in a decreasing order of their sales rankings. Similarly, Steam, one of the largest video game platforms in the world, displays an ordered list of top sellers (Figure 1, right panel). Neither Amazon nor Steam, however, reveals sales volume for any product, regardless of the product's top selling status. In contrast, Taobao.com, a major Chinese online shopping platform, discloses each product's "number of units sold in the last seven days" on its bestselling chart of a product category (see an example in Figure 2, left panel). In addition, different from Amazon and Steam, which display bestsellers for either predefined product categories or all products, Taobao allows sorting product search results by units sold, essentially enabling shoppers to create their own bestseller lists with search keywords, as exemplified in the right panel of Figure 2. Similar practices are adopted by JD.com, Taobao's key competitor in China.

Given the prevalence of bestseller lists and the leading platforms' diverse practices, several groups of interrelated questions naturally arise: (i) How informative is a bestseller list about product values? Are products ranked higher on the list always of higher value, and if not, why? (ii) How does a bestseller list of varying granularity impact late-arriving consumers' beliefs about product values and, as a result, influence their product search and purchasing decisions? (iii) Should a revenue-maximizing platform release bestseller information to consumers, and if so, how? In particular, is finer information necessarily better, and which form, sales ranking only or ranking with sales volume, should the platform adopt? How should the platform time its information provision? (iv) From the consumers' perspective, how does bestseller information of varying granularity affect average value of a purchased good, expected search cost incurred before purchase, and overall consumer

Figure 1. (Color online) Bestseller Lists on Amazon.com (Left) and Store.steampowered.com (Right)

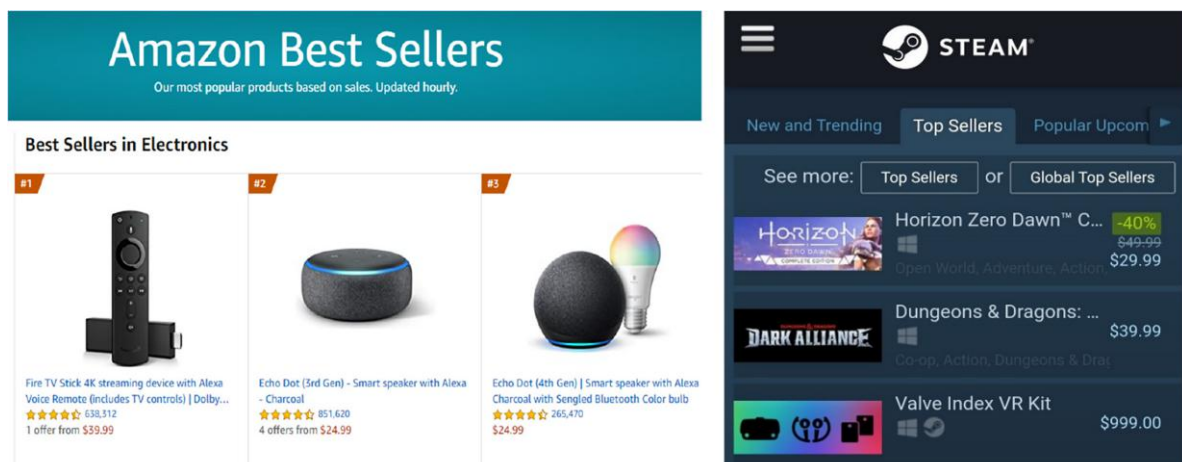
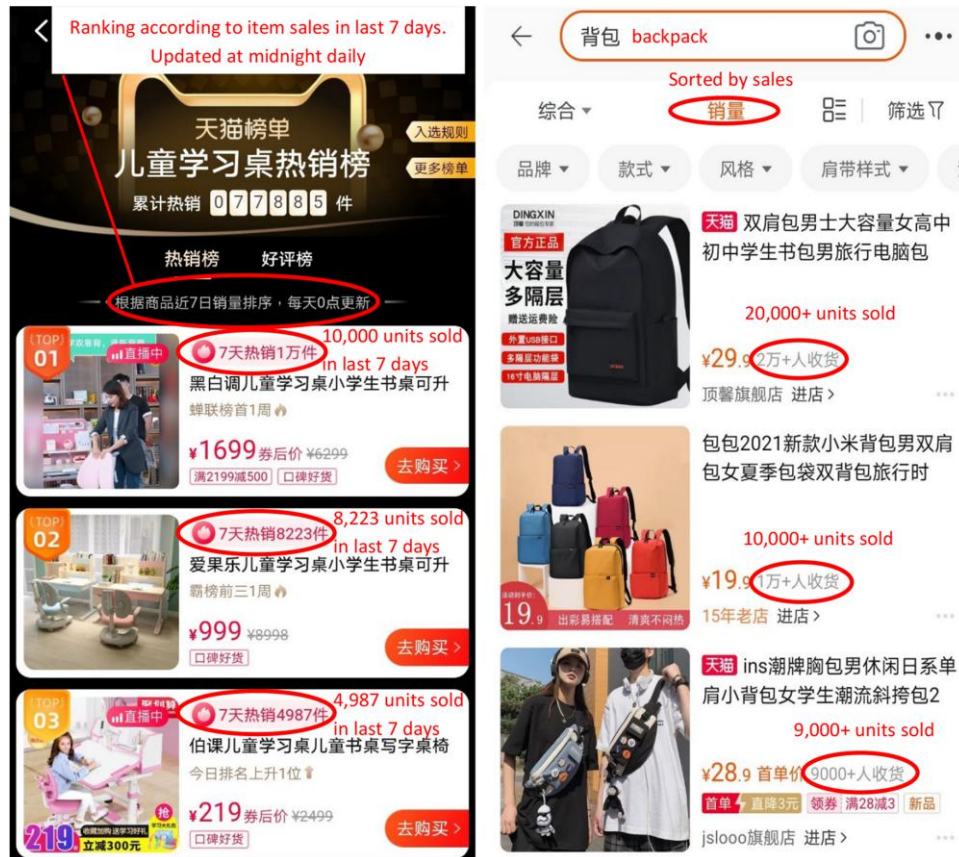


Figure 2. (Color online) Taobao.com: Bestselling Chart for Kids' Study Desk (Left) and Search Results for the Keyword "Backpack" Sorted by Sales (Right)



Note. English translation is added by the authors.

surplus? How do these results inform a platform that, instead of maximizing revenues, adopts a consumer-centric goal of maximizing consumer welfare? Should such a platform provide bestseller information, and if so, what is its most preferred level of information granularity?

In general, answers to these questions may be influenced by a multitude of factors, such as market, competition, and regulatory policies. Our work focuses on bestseller list generation through consumers' private and costly acquisition of product information (i.e., consumer search) and, thus, centers around factors governing consumers' product search and learning. Key factors identified in our study include consumers' search cost heterogeneity, product value perception, and population size.

Specifically, we analyze a model of social learning with consumer search and search-generated sales information. Consumers arrive in two cohorts (early and late) to a platform on which two products are listed for sales. Each consumer chooses which product (if any) to purchase under uncertainty about the products' values, which follow a bivalued prior distribution. Before making the decision, a consumer has the option

to sequentially search the products. A product search perfectly reveals the value of the searched product to the consumer and is yet associated with a positive search cost. Consumers are heterogenous in search cost and independently decide which product(s), if any, to search and to buy. A major difference between the early and late consumers is that the latter can access, at the platform's discretion, information about the product sales to the early consumers. We evaluate sales information of two levels of granularity: sales ranking and sales volume. The platform chooses whether to disclose sales information and, if so, the granularity level and timing to publicize the information (the latter determines the sizes of the cohorts) to maximize its total expected sales.

Our model and analysis lead to the following important findings and insights.

For question (i), we find that search-generated bestseller lists are informative about product values because higher value products are more likely to be ranked higher than lower value ones. Nevertheless, bestseller lists can also be misleading as a higher ranked product may be of lower value (Proposition 1). A key determinant of the bestsellers' informativeness is the number of

early consumers with low search costs, who can afford to search both products and, subsequently, always purchase a high-value product if there is any. On the other hand, the early consumers who purchase a randomly chosen product without researching both products contribute to the noise in the bestseller information.

Regarding impact of bestseller lists on consumer search and purchase (question (ii)), we show that, despite the noise in bestseller information, it is rational for the late-arriving consumers to first search the higher ranked product, which is believed to be of high value with a higher chance (Proposition 2). Thus, the consumers may be misguided to investigate first the lower value product and, consequently, overlook the higher value one. This finding appears similar to and yet is fundamentally different from the well-known information cascade in the observational learning literature (Banerjee 1992), as it is driven by the positive and heterogeneous cost of consumers' private information acquisition.

Furthermore, a bestseller list revealing only sales rankings always boosts the late-arriving consumers' willingness to search and purchase because it enhances their confidence about discovering high value in first search (Proposition 3). In contrast, sales information of greater detail, about numbers of units sold, may dampen consumers' motivation to purchase (Proposition 4). In particular, a small difference in sales may indicate that most probably the products are of identical value. Thus, if consumers hold a pessimistic view about high product value initially, a small sales difference strengthens their belief about both products being of low value and, consequently, disincentivizes them to purchase either product (Proposition 5). This inference effect, which we refer to as reinforcement by homogeneity, is unique to multiproduct Bayesian learning. For a similar reason, compared with disclosing sales ranking only, revealing a small sales difference is less effective in promoting future purchases because a small sales gap suggests that the bestseller may have prevailed over the other product by pure luck.

In terms of the platform's information provision strategy (question (iii)), we find that, whereas providing the information is advantageous to the platform, information of higher granularity may not be its preferred choice. Specifically, disclosure of sales ranking always benefits the platform, whereas revealing sales volume may backfire (Proposition 6). By comparing the expected sales under ranking and volume information, we show that, if the cumulative distribution function of the search cost distribution is concave, providing bestseller ranking outperforms disclosing sales volume from the platform's perspective. We also prove that the percentage sales difference between volume and ranking information can be arbitrarily high (Proposition 7).

In addition to information granularity, the platform can leverage the timing of information provision by striking a balance between information exploration and exploitation. Under a search cost distribution empirically developed in the consumer search literature, we find that the platform should extend the stage of information exploration as the consumer population grows and ranking information is more likely to be preferred by the platform under either a larger consumer population or a stochastically higher product value. These results, together with that on search cost distributions, suggest plausible explanations for the leading platforms' diverse practices of bestseller information provision.

From the consumers' viewpoint (question (iv)), even though the bestseller information is noisy, it never hurts the consumers on expectation, and information of higher granularity benefits the consumers more (Proposition 10). Thus, the bestseller information that consumers favor the most is of the highest granularity. In particular, sales volume is more informative about product values than sales ranking as volume allows consumers to learn about the joint distribution of the product values, whereas ranking merely enhances the higher ranked product's expected value relative to the lower ranked one's. Our findings indicate a potential misalignment in platforms' and consumers' preferences over information granularity, suggesting that platforms may withhold information at the cost of consumers. On the other hand, platforms aiming to maximize consumer surplus should strive to enhance bestseller information transparency on their marketplaces.

Whereas the aggregate consumer surplus increases in sales information granularity, sales information (of a higher granularity) does not always lead to an improvement in a consumer's expected purchased value or a saving in the consumer's expected search cost (Propositions 8 and 9). The latter implies that public learning may not be substitutable to private learning.

Last but not the least, we confirm our main findings in various extensions on consumer behavior, generalized product settings, and information provision strategies. We derive insights, among others, regarding how, when, and why strategic consumer waiting for bestseller information can be advantageous to the platform and why information personalization may benefit the platform but not the consumers. We also assess how product asymmetry, number of products, extra consumer heterogeneity, and more frequent information updating affect the informativeness of bestseller lists and illustrate the impact of product sellers' pricing flexibility on bestseller information provision.

2. Literature Review

The work bridges three literature streams: consumer search, social learning, and information provision.

Consumer search, in its essence, is consumers' costly information acquisition. Main research questions examined in the extant literature of consumer search include consumers' optimal search strategies and their effects on firms' decisions. Many studies in this stream assume consumer homogeneity in search cost (e.g., Compiani et al. 2024). Particularly relevant to our paper are the prior studies considering consumers' search cost heterogeneity. For example, Hong and Shum (2006) and Moraga-González et al. (2013) provide empirical evidence of search cost heterogeneity and estimations of search cost distributions. Derakhshan et al. (2022) model online platforms on which search costs are both heterogeneous among consumers and dependent on product ranking. Among the theoretical studies that reveal the important roles of search cost heterogeneity in determining firms' optimal selling strategies, Moraga-González et al. (2017a, 2017b) characterize competitive pricing equilibrium in presence of consumer search and show that the shape of the search cost distribution (e.g., dispersed or concentrated) may lead to qualitative changes in the pricing equilibrium. Jin et al. (2023) find that artificial sales inflation by fake orders may increase consumer welfare, and this effect can be strengthened by search cost heterogeneity. Our work complements the consumer search literature by explicitly modeling and analyzing consumer search as a key determinant of how a bestseller list is generated and how it impacts future sales. We demonstrate that search cost heterogeneity leads to early arriving consumers' diverse search and purchasing behavior, which determines the informativeness of a bestseller list to late arriving consumers.

The literature of social learning describes and investigates the processes through which consumers learn from their peers. Different from the studies on social learning via consumer-generated information (e.g., consumer reviews or shared observations as in, e.g., Papanastasiou and Savva 2016, Yu et al. 2016, Wang and Hu 2020), our work is akin to the literature stream on observational learning, in which consumers learn from others' actions, for example, purchasing. In particular, a consumer's purchasing action can be informative to other consumers as it may reflect the knowledge that the consumer has acquired from private channels. Many studies on observation learning assume that such private information acquisition is costless. See, for example, Banerjee (1992) and Callander and Hörner (2009). In contrast, our model is more closely related to those capturing consumers' costs during their private information acquisition. A major modeling difference arises because of such costs: when private information is costly, consumers rationally choose whether to acquire it, and thus, the informativeness of their actions becomes endogenous. Under costly private information, observational learning is more complicated: late

arriving consumers need to deduce early arrivals' decision rules to anticipate actions taken with private knowledge and those without, as these actions differ in the information that they imply.

A few recent papers consider observational learning under costly private information. Hendricks et al. (2012) analyze a single-product model in which each consumer can access two private signals: a free and noisy signal and a costly and perfect signal. Based on the free signal and aggregate purchasing records of previous consumers, each consumer makes a decision on whether to acquire the costly signal and purchase the product. Papanastasiou (2020) models consumers' decisions on whether to perform a costly inspection of a news article to verify its validity and whether to share the news to other consumers. Both Hendricks et al. (2012) and Papanastasiou (2020) assume a single opportunity of private information acquisition. Different from these works, our two-product model captures consumers' decisions on which product to search and which to buy and, particularly, incorporates their option of a second search. We show that the second search plays a central role in determining the informativeness of a bestseller list. We also find that a bestseller list imposes opposite effects on the first and second searches: it promotes the former and discourages the latter. In a similar vein, Mueller-Frank and Pai (2016) and Garcia and Shlegia (2018) analyze agents'/consumers' sequential search of multiple actions/products in conjunction with observational learning. In contrast, we highlight the importance of search cost heterogeneity and focus on implications of learning through bestseller lists.

Furthermore, a major departure of our work from the extant observational learning literature is that, instead of assuming consumers' direct observation of others' actions (in the form of either entire history or aggregate records), we consider the practical situation in which consumers can only access such information through a self-interested intermediary, that is, the platform, which privately observes the purchase history and optimizes its information disclosure to maximize expected sales.

Our paper is also closely related to the literature on strategic information provision by firms to consumers. Some work in this vein examines sellers' provision of product information, such as Branco et al. (2016), Liu et al. (2019), and Calvo et al. (2023). Among these studies, both Branco et al. (2016) and Liu et al. (2019) capture the role of information provision in reducing or eliminating consumers' search costs and reveal that a full disclosure of product information can be suboptimal for a seller, and these are relevant to our work. Different from product information, however, a bestseller list is about aggregate purchase history and, hence, the insights in these studies do not apply to the provision problem in this paper. In settings without

consumer search, Hu et al. (2013, 2020) analyze firms' information provision regarding early arriving consumers' actions, which facilitates late arriving consumers' inference about market size and the success rate of a group buying deal, respectively. Our work differs by focusing on learning about product values. A platform's strategic provision of sales information for product value learning is recently investigated by Küçükgül et al. (2022). The authors model the platform as an information designer, which is privately informed about up-to-date sales history and develops accordingly a message to each consumer. They consider a single-product setting in which each consumer observes, in addition to the platform's message, a noisy and costless private signal. Our work analyzes consumers' choice between two products and differs from Küçükgül et al. (2022) by focusing on bestseller information, which gains its value and affects future sales through consumers' sequential and costly acquisition of private product knowledge.

From a broader perspective, this paper is also related to an emerging stream of literature in operations management on information design. Studies in this stream focus on optimizing information senders' signaling strategies under various operational contexts, such as learning about content accuracy (Candogan and Drakopoulos 2020), warning against a recurring disaster (Alizamir et al. 2020), inducing consumers' early purchases (Drakopoulos et al. 2020), investing on crowdfunding platforms (Glazer et al. 2021), and reducing congestion in a service system (Anunrojwong et al. 2023). Our paper evaluates the effectiveness of signaling through bestseller lists to facilitate consumers' product search and purchasing decisions on e-commerce platforms. We focus on two types of bestseller information, sales ranking and sales volume, which are prevalent in practice and, as revealed in the paper, optimal within a general class of sales signals under some conditions.

To summarize, our work connects and contributes to the aforementioned literature streams in three major aspects. First, we embed consumers' sequential search in a social learning framework and demonstrate its central role in the process of generating, disseminating, and learning through bestseller information. Second, by comparing the learning outcome under sales ranking and sales volume, we derive insights into how the granularity of historical sales information impacts consumers' search and purchasing decisions as well as its sales/welfare implications to the platform/consumers. Third, we evaluate various provision strategies of the platform and reveal interesting interplay between the granularity and timing of its bestseller information disclosure. The insights also contribute toward a more comprehensive understanding of the interaction between bestseller information and various

operational factors as well as of platforms' trade-off in determining bestseller information provision.

3. Model

3.1. The Products

Consider two new products, labeled by $i \in \{1, 2\}$, being listed on a platform for sales to a pool of consumers. The products are ex ante identical and, thus, are offered for sale at a common price f (normalized to zero without loss of generality). Specifically, the platform and the consumers share a common prior belief that, by consuming each product, a consumer enjoys a high value, u_h , with probability p_h and a low value, u_l , with probability $p_l := 1 - p_h$, where $u_h > u_l > 0$ and $p_h \in (0, 1)$.² Let $\Delta := u_h - u_l$. The actual values of the products, denoted by u_1 and u_2 , are determined by nature, which randomly and independently draws a value for each product from the aforementioned binary distribution. Hence, there are four possible combinations for the products' values (u_1, u_2) : (u_h, u_h) , (u_h, u_l) , (u_l, u_h) , and (u_l, u_l) , occurring with probabilities p_h^2 , $p_h p_l$, $p_l p_h$, and p_l^2 , respectively. The actual value of a product is unobservable to a consumer until the consumer either purchases or searches the product (as elaborated below). The values are unobservable to the platform throughout the selling horizon, as the platform is a sales (and information) intermediary and uninvolved in physical handling of the products during order fulfillment. For convenience, we also refer to a product of high (low) value as a high (low) type.

3.2. The Consumers

A total of n consumers, indexed by $j \in N := \{1, 2, \dots, n\}$, sequentially arrive to the platform.³ Each consumer has a unit demand and decides whether and, if so, which product to buy. As aforementioned, consumers are uncertain about the products' values and hold a common belief that $\mathbb{P}(u_i = u_h) = p_h$ and $\mathbb{P}(u_i = u_l) = p_l$, $i = 1, 2$. Before making a purchase, each consumer can choose to perform a costly investigation of the product's value by, for example, perusing product descriptions, watching demo videos, and visiting physical showrooms to assess the product in person. We refer to such an investigation as a search of the product. Searching a product fully reveals the product's value to the consumer, who then chooses to either purchase the searched product or, alternatively, continue to search the other product and decide which to buy afterward. That is, a consumer can conduct up to two searches before making a purchase. For every product search, a consumer incurs a search cost, denoted by s . Consumers are heterogeneous in their search costs, which follow a general distribution on $[0, \bar{s})$ with cumulative distribution function $F(\cdot)$. That is, among any m randomly selected consumers ($m \leq n$),

the number of consumers whose search cost is less than x equals to $mF(x)$.⁴ For analytical convenience, we assume that the upper bound of the distribution support, $\bar{s}$, is sufficiently large (it suffices to assume $\bar{s} > u_h$) such that the search cost of the marginal consumer (i.e., the consumer who is indifferent about whether to search) is always in the interior of the distribution support. The search cost distribution $F(\cdot)$ is public knowledge, whereas each consumer is privately informed about the consumer's own search cost. Following prior studies on search (e.g., Hendricks et al. 2012, Garcia and Shelegia 2018), we assume that a consumer cannot purchase a product until the consumer searches it. We also assume that consumers' reservation utility is zero (e.g., Boleslavsky et al. 2017, Liu et al. 2019). We relax these two assumptions and show that our main findings are robust (see Section 7.3.2). For tie breaking, assume that a consumer does not search a product if the probability of the consumer's utility being improved after the search (and a subsequent search if possible) is zero.

Product search represents a private channel for consumers to acquire product information, and it is up to each individual consumer to decide whether to pursue such information. In addition to product search, consumers arriving late (as explained below) can access some public information about the purchasing decisions made by consumers arriving early. Specifically, the information can be in the form of either ranking or numbers of units sold of the products, accumulated from the beginning of the selling horizon to the moment that a certain number of consumers have arrived to the platform. The form of such public information (i.e., sales ranking or volume) as well as when it is publicized (i.e., after how many consumers have arrived) are decisions of the platform. From the sales information the late arriving consumers can gain some insights into the search outcomes of the early arrivals, which, in turn, lead to their belief updating about product values. Accordingly, the late arrivals make decisions on product search. All consumers are rational and aim to maximize their own expected utilities.

3.3. The Platform

As a common practice for digital commerce, the platform charges a commission fee for transactions on its website. Because the retail price is beyond the platform's control and identical for the two products (as they are ex ante homogenous), the platform's objective is to maximize the total product sales.⁵ Aided by web analytics tools and techniques, the platform is able to track consumer purchases. To facilitate social learning, the platform can release information about purchases made by early arriving consumers. Because of legal and privacy considerations, such information

must be on aggregate levels. We examine and compare three information provision strategies of the platform: no information, sales ranking, and sales volume.⁶ In the meanwhile, the platform also chooses the timing for its information provision. Specifically, it determines a critical number, n_0 , and publicizes the sales information after n_0 consumers have arrived.⁷ We assume that both the granularity and timing of the platform's information provision are determined up front. The release of the information naturally partitions the selling horizon into two phases: an early phase in which sales information is not available and a late phase in which it is. Hereafter, we refer to these two phases also as the first and second periods, respectively, and consumers arriving in the two phases as the early and late consumers, respectively.

3.4. The Game

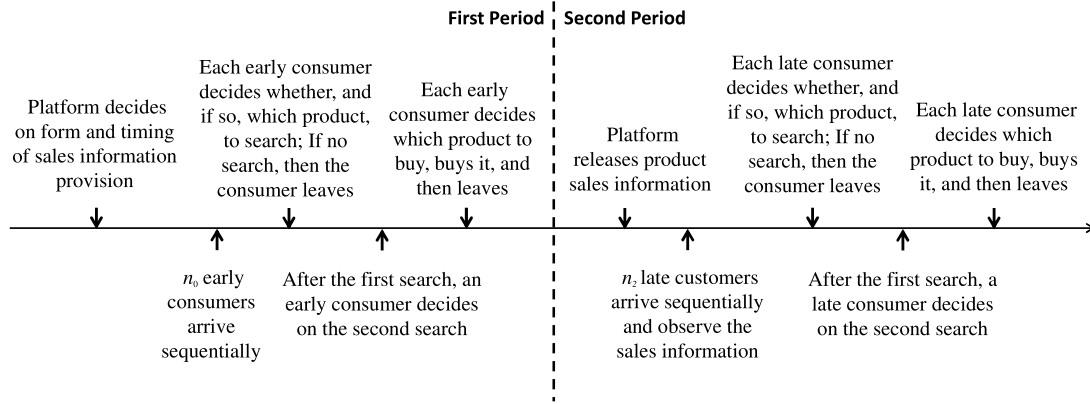
The game chronology is shown in Figure 3, in which $n_2 := n - n_0$ represents the number of consumers arriving in the second period. This is a two-period Bayesian learning model, of which a unique feature is social learning of product information endogenously generated by early consumers' sequential product search and strategically controlled by the self-interested platform.

In what follows, we first analyze the consumers' decisions, fixing the timing and type of the sales information. We start from the early consumers' search and purchasing decisions, which determine the first period product sales. Subsequently, we characterize how the late consumers learn about product values from the sales information (if any) and make their own search and purchasing decisions. On the basis of these consumer analyses, we then analyze the platform's information provision strategies.

4. Consumers' Optimal Search and Purchasing Strategies

4.1. First Period (Information Generation)

Without any public information in the first period, the early consumers follow the common prior belief to decide whether to search a product, and those with a low search cost choose to do so. Because the two products are ex ante identical, a searching consumer randomly picks a product and ascertains its value. If the revealed value is u_h , a further search cannot result in a higher value, and the consumer buys the searched product. If, however, the searched product is of low value, then a second search may be worthwhile if the consumer's search cost is very low. As the search cost is sunk and $u_l > 0$, all the consumers who have performed the first search purchase a product. Formally, a first period consumer with search cost s makes a first search (and purchases a product) if

Figure 3. Sequence of Events

and only if

$$p_h u_h + p_l \max(u_l, p_h u_h + p_l u_l - s) - s \geq 0, \quad (1)$$

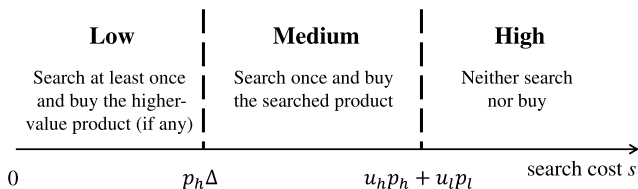
and conditional on a low value revealed in the first search, the consumer makes a second search if and only if $p_h u_h + p_l u_l - s \geq u_l$. Lemma 1 characterizes the first period consumers' purchasing choices as induced by their search behavior. Figure 4 provides an illustration.

Lemma 1 (Consumers' Purchasing Choices in the First Period).

i. In the first period, consumers with search cost $s \leq u_h p_h + u_l p_l$ purchase a product and consumers with higher search cost leave without buying. That is, the total sales of the two products in the first period is $n_1 := \lfloor n_0 F(u_h p_h + u_l p_l) \rfloor$, where $\lfloor x \rfloor$ represents the largest integer no greater than x .

ii. If the products' values are identical (i.e., either $u_1 = u_2 = u_h$ or $u_1 = u_2 = u_l$), a consumer who makes a purchase buys either product with equal probability. If the products' values are different (i.e., $u_i = u_h$ and $u_{3-i} = u_l$, $i \in \{1, 2\}$), a consumer with search cost $s \leq p_h \Delta$ purchases the product with high value, and one with search cost $s \in (p_h \Delta, u_h p_h + u_l p_l)$ purchases the first product that the consumer searches.

Lemma 1(i) shows that a first period consumer conducts a first search as well as buys a product if and only if the consumer's search cost is no greater than either product's expected value evaluated with the prior belief, that is, $u_h p_h + u_l p_l$. Thus, independent of the actual product values, the two products' sales in the first period add up to a constant. It is worth noting

Figure 4. Search and Purchasing Decisions of First Period Consumers

that these results coincide with their counterparts in a hypothetical scenario in which each consumer has only one search opportunity (in this case, both search and purchasing are completely random and, thus, sales cannot generate any useful information about product value). In other words, the option of a second search does not impact on consumers' willingness to do a first search: although the option is attractive to some consumers and boosts their expected payoff, these consumers have low search costs and perform a first search independent of the option; those consumers with high search costs would not search regardless of the option.

Whereas the opportunity to search the other product has no implication for whether an early arriving consumer purchases, it leads to heterogeneity in consumers' purchasing choices, and its impact differs by the products' values, as in Lemma 1(ii). Thus, the second search option creates dependence of products' sales on their values and, consequently, allows social learning through sales information.

Specifically, conditional on the products being of identical values, a consumer's purchasing choice is purely random and independent of whether the consumer conducts a second search. To elaborate, if both values are high, a second search never occurs, and each consumer purchases the product that the consumer randomly picks in the consumer's first search (if the consumer chooses to search); if both values are low, the consumers with medium search costs buy either product through the first search, whereas those with low search costs randomly purchase a product after performing the second search. Hence, in this case, the products' sales are identically distributed, implying that the products have equal chances to become the bestseller.

Lemma 1(ii) also describes consumers' purchasing choices if the products' values differ from each other. As consumers with medium search costs search only once and purchase the searched product regardless of

its value, their purchases do not convey any information about product values. Instead, their purchases of a randomly picked product give rise to the possibility of the low-value product overselling the high-value one, in which case the bestseller information misleads the late consumers. In contrast, consumers with low search costs inspect both products and purchase the one with high value. Hence, their purchases are indicative of product values. That is, purchases by the low-cost consumers generate useful product information, whereas others' purchases add noise to that information. In this case, the high-value product is most likely the bestseller, but with a small probability, the opposite occurs because of the noise brought by the consumers who search only once.

Based on Lemma 1, the following proposition specifies first period sales distributions, which reflect how the bestseller information is shaped by early arriving consumers' search and purchasing decisions.

Proposition 1. Define $m := \lfloor n_0 F(p_h \Delta) \rfloor$, and for $x \in [0, n_1]$, $g_s(x) := \text{Binomial}(x, n_1, 1/2)$, $g_a(x) := \text{Binomial}(x - m, n_1 - m, 1/2)$ if $x \geq m$, and 0 if $x < m$, where $\text{Binomial}(x, y, p)$ is the probability that, among y independent trials, x of them succeed, where the probability of success is p . Let $G_s(x)$ and $G_a(x)$ be the cumulative distribution functions corresponding to $g_s(x)$ and $g_a(x)$, respectively. Let $\bar{G}_s(x) := 1 - G_s(x)$ and $\bar{G}_a(x) := 1 - G_a(x)$.

- i. If the two products' values are identical, the sales of either product follows distribution $G_s(x)$.
- ii. If the two products' values are different, the sales of the high-value (respectively, low-value) product follows distribution $G_a(x)$ ($\bar{G}_a(n_1 - x)$). Furthermore, the probability that the higher value product has higher sales is less than one if $n_1 \geq 2m$.

Proposition 1 highlights the fact that a higher value product (if any) is not necessarily sold more. In particular, when only one product is of high value, the high-value product is purchased by the consumers who search both products (with a total mass of m) as well as by the consumers who, among those searching only once (with a total mass of $n_1 - m$), happen to pick the high-value product during their random draws from the two products. If $n_1 \geq 2m$, in the worst case scenario in which all of the $n_1 - m$ consumers draw the low-value product, the low-value product turns out to be the bestseller.

4.2. Second Period (Social Learning)

Different from the first period consumers, who can acquire product information only through private search, the second period consumers can learn from an additional source, that is, through the first period sales information revealed by the platform. In this subsection, we examine consumer learning under three types of sales information (no information, sales ranking, and sales volume).

Whereas the products are ex ante homogenous, they are no longer so to the second period consumers, who can differentiate the products according to their first period sales. Let X_i be the sales for product $i = 1, 2$, where $X_1 + X_2 = n_1$. We refer to the product with higher (lower) sales as product i^* ($-i^*$). Denote product i^* 's first period sales by X_{i^*} and its realization by $x \in [n_1/2, n_1]$. The product with higher sales is ranked higher on the bestseller list. In the cases that n_1 is even and the first period sales are equal (i.e., $x = n_1/2$), we assume that it is equally likely for either product to be ranked higher than the other. Let $t \in \{\phi, r, v\}$ denote the type of information with ϕ , r , and v corresponding to no information, sales ranking, and sales volume, respectively. Let π_1^t denote a consumer's posterior belief that the higher ranked product is of high value upon observing information t and before any search. Because the late consumers observe which product is the bestseller and the products are symmetric ex ante, we define $\pi_1^\phi = p_h$, $\pi_1^r = \Pr[u_{i^*} = u_h | i^* = 1]$, and $\pi_1^v(x) = \Pr[u_{i^*} = u_h | X_{i^*} = x, i^* = 1], x \geq n_1/2$. We also define v_1^t as a consumer's posterior belief that the product with lower sales is of high value upon the consumer observing information t and before conducting any product search. That is, $v_1^\phi = p_h$, $v_1^r = \Pr[u_{-i^*} = u_h | i^* = 1]$, and $v_1^v(x) = \Pr[u_{-i^*} = u_h | X_{i^*} = x, i^* = 1], x \geq n_1/2$.

By comparing π_1^t and v_1^t , we first prove in Proposition 2 that, if a consumer decides to search a product, the consumer should first search the bestseller.

Proposition 2. Under either ranking or volume information, if a consumer finds it worthwhile to search, then it is optimal for the consumer to first search product i^* .

Proposition 2 is driven by consumers' rational belief that the higher ranked product is more likely to be of high value than the lower ranked one (i.e., $\pi_1^t \geq v_1^t$). This is because, conditional on the product values differing from each other, the sales of the higher value product is stochastically greater as those early consumers who search twice purchase a high-value product if any. Therefore, despite the noise in bestseller information (i.e., the higher ranked product may be of lower value), it is optimal for consumers to follow the information in their first search. This result is empirically corroborated in Fu (2020), which finds that consumers are more likely to first click products with higher sales volume.

Similar to the situation in the first period, if a consumer's first search reveals a product of high value, the consumer buys the product. If, however, the searched product is of low value, the consumer updates the belief about the remaining product and decides whether to search it. Denote by π_2^t a consumer's posterior belief about the lower ranked product being high type under information t and conditional on the higher ranked product being of low value. That is,

$\pi_2^\phi = p_h$, $\pi_2^r = \Pr[u_{-i^*} = u_h | u_{i^*} = u_l, i^* = 1]$, and $\pi_2^v(x) = \Pr[u_{-i^*} = u_h | X_{i^*} = x, u_{i^*} = u_l, i^* = 1], x \geq n_1/2$.

Lemma 2 shows that, for a given information type t , the posteriors π_1^t , v_1^t , and π_2^t are nicely ordered.

Lemma 2 *The posteriors satisfy the following inequalities: $\pi_1^r \geq v_1^r \geq \pi_2^r$, and $\pi_1^v(x) \geq v_1^v(x) \geq \pi_2^v(x)$, $\forall x \geq n_1/2$.*

In addition to comparing the beliefs for first search ($\pi_1^t \geq v_1^t$), Lemma 2 also reveals that, after a consumer realizes, through the consumer's first search, that the higher ranked product is a low type, the consumer becomes even more skeptical about deriving a high value from the lower ranked product than before the first search (i.e., $v_1^t \geq \pi_2^t$), as a low type outselling a high type rarely occurs. The lemma also implies that a second period consumer is more optimistic about discovering a high-value product in the consumer's first search (of the higher ranked product) than in the consumer's second search (of the lower ranked product), that is, $\pi_1^t \geq \pi_2^t$.

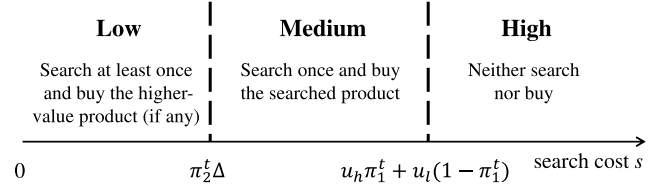
Lemma 2 suggests two implications of social learning for private learning. First, social learning converts consumers' random product searches to directional searches, starting from the product more promising for high value. Second, sales information leads to product ordering and creates interdependence between product values: although, in the first period, the outcome of a first search has no impact on the other product's expected value, in the second period, discovering a low type in the first search renders a consumer more doubtful about finding a high type by the second search.

Regarding the decision on whether to search, anticipating the search–search option, a second period consumer makes a first search (and purchases a product) if and only if $\pi_1^t u_h + (1 - \pi_1^t) \max(u_l, \pi_2^t u_h + (1 - \pi_2^t) u_l - s) \geq 0$, and conditional on a low-value product revealed in the first search, the consumer makes a second search if and only if $\pi_2^t u_h + (1 - \pi_2^t) u_l - s \geq u_l$. Lemma 2 and Proposition 2 jointly imply that the second period consumers' optimal searching and purchasing strategies are identical, structure-wise, to those for the first period consumers. These are characterized in Lemma 3 and illustrated in Figure 5.

Lemma 3 (Consumers' Purchasing Choices in the Second Period).

- i. Consumers with search cost $s \leq u_h \pi_1^t + u_l(1 - \pi_1^t)$ purchase a product, and consumers with higher search cost leave without buying. That is, the total sales of the two products in the second period is $n_2 F(u_h \pi_1^t + u_l(1 - \pi_1^t))$.
- ii. If a consumer's first search reveals a low type, the consumer performs a second search if and only if the consumer's search cost is low (i.e., $s \leq \pi_2^t \Delta$). Thus, if the two products' values are different, a consumer with search cost $s \leq \pi_2^t \Delta$ purchases the product with high value, whereas a consumer with search cost $s \in (\pi_2^t \Delta, u_h \pi_1^t + u_l(1 - \pi_1^t))$ purchases the first product that the consumer searches.

Figure 5. Search and Purchasing Decisions of Second Period Consumers



By Lemma 3, the impact of sales information on consumers' searches and purchases is solely through its impact on the two posteriors π_1^t and π_2^t , which we examine next.

4.3. Impact of Sales Information on Consumers' Posterior Beliefs

4.3.1. Ranking Information vs. No Information. We start from comparing the second period consumers' posterior beliefs under ranking information with those without any sales information.

Proposition 3 (Impact of Ranking Information on Posteriors). $\pi_1^r \geq \pi_1^\phi = p_h \geq v_1^r$, $\pi_2^r \leq \pi_2^\phi = p_h$. That is, under ranking information, the belief that product i^* ($-i^*$) has high value is higher (lower) than the prior (i.e., without ranking information), and conditional on i^* being of low value, the belief that product $-i^*$ has high value is lower than the prior (i.e., without ranking information).

The logic behind Proposition 3 is as follows. Recall from Section 4.1 that, if the products differ in value, the higher value product is more likely the bestseller; whereas if the products share the same values, their sales are identically distributed. Thus, consumers associate a higher sales ranking with a greater expected value. In particular, the posterior belief for the higher ranked product being of high value is higher than the prior, and that for the lower ranked product is lower than the prior. For a similar reason, a low-value product is less likely to achieve a top sales ranking when the other product is of high value than when the other product is a low type. Hence, ranking information lowers consumers' confidence about finding high value in their second search.

To further elucidate the effect of ranking information, we decompose the posterior π_1^r as follows:

$$\begin{aligned} \pi_1^r &= \Pr[u_1 = u_h | X_1 \geq \frac{n_1}{2}] \\ &= \Pr[u_1 = u_h | u_1 \neq u_2, X_1 \geq \frac{n_1}{2}] \Pr[u_1 \neq u_2 | X_1 \geq \frac{n_1}{2}] \\ &\quad + \Pr[u_1 = u_h | u_1 = u_2, X_1 \geq \frac{n_1}{2}] \Pr[u_1 = u_2 | X_1 \geq \frac{n_1}{2}], \end{aligned}$$

where the first equality follows from symmetry of the two products.

Lemma 4 (Impact of Sales Ranking on Posterior Beliefs: Decomposition). For $i = 1, 2$,

i. $\Pr[u_i = u_h | u_1 \neq u_2, X_i \geq n_1/2] \geq \Pr[u_i = u_h | u_1 \neq u_2]$
and $\Pr[u_i = u_h | u_1 = u_2, X_i \geq n_1/2] = \Pr[u_i = u_h | u_1 = u_2]$.

ii. $\Pr[u_1 \neq u_2 | X_i \geq n_1/2] = \Pr[u_1 \neq u_2]$ and $\Pr[u_1 = u_2 | X_i \geq n_1/2] = \Pr[u_1 = u_2]$.

Lemma 4 details the effects of ranking information on the probability of the bestseller being of high value conditional on product values as well as on the beliefs for product values. The first inequality in part (i) of the lemma reveals the fundamental reason why sales ranking is informative. Specifically, because of the early consumers' search and purchasing behavior, the higher ranked product is more likely to be the high-value one if the products differ in value. If, however, the product values are equal, sales ranking is uninformative because, in this case, all the early consumers randomly pick a product for purchase. It is important to note that consumers cannot derive any useful information from sales ranking as to whether the products are of different or identical values (part (ii) of the lemma). That is, sales ranking does not convey any informative signal about the joint distribution of product values. Essentially, this is because the products are ex ante homogenous, and thus, the probability of a randomly chosen product being the bestseller is independent of whether the values differ from each other, that is, $\Pr[X_i \geq n_1/2 | u_1 = u_2] = \Pr[X_i \geq n_1/2 | u_1 \neq u_2] = \Pr[X_i \geq n_1/2] = 1/2, i = 1, 2$. Hence, knowing sales ranking does not alter consumers' belief about whether both, either, or neither product is of high value. This result is in stark contrast with that under sales volume information as we show next.

4.3.2. Volume Information vs. No Information. Proposition 4 characterizes the impact of volume information on consumers' posteriors.

Proposition 4 (Impact of Volume Information on Posteriors).

- i. $\pi_2^v(x) \leq p_h$ for all $x \geq n_1/2$.
- ii. If $p_h \geq 1/2$, $\pi_1^v(x) \geq p_h$ for any $x \geq n_1/2$; otherwise, $\pi_1^v(x) \geq p_h$ if and only if x is sufficiently high.

Proposition 4(i) proves that, similar to ranking information, volume information lowers the posterior governing consumers' second search decision (i.e., π_2) because a low-value product is more likely to be the bestseller when both products' values are low than when only one of them is. Nevertheless, the effect of volume information on the posterior preceding the first search, π_1 , is drastically different from that of ranking information. Specifically, whereas ranking information always enhances consumers' confidence about realizing a high value in the bestselling product,

volume information may dampen it when the difference between the two products' first period sales is small (or, equivalently, when the bestseller's sales, x , is low) as in part (ii) of the proposition. In other words, volume information can disincentivize a consumer's first search and reduce the chance of the consumer purchasing either product.

To build intuition for this result, we first examine an extreme case of $x = n_1/2$, that is, when the products' first period sales are equal to each other. As shown in the lemma below, the posterior for the bestseller being a high type, upon observing equal product sales, can be strictly lower than the prior.

Lemma 5 (Impact of Volume Information When First Period Sales Are Equal). $\pi_1^v(n_1/2) > p_h$ if $p_h > 1/2$, $\pi_1^v(n_1/2) = p_h$ if $p_h = 1/2$, and $\pi_1^v(n_1/2) < p_h$ if $p_h < 1/2$.

Lemma 5 is driven by a fundamental effect in multi-product learning, which we refer to as (belief) reinforcement by (value) homogeneity. Specifically, once a consumer is convinced that the products are of equal value, the consumer becomes more optimistic about deriving a high value from either product if and only if the consumer holds a stronger belief, ex ante, for the high value than the consumer does for the low value. Formally, conditional on equal product values, the probability that either product is of high value is greater (less) than the prior of the high value if the latter is greater (less) than a half, that is, for $i \in \{1, 2\}$, $\Pr[u_i = u_h | u_1 = u_2] = p_h^2 / (p_h^2 + p_l^2) > (\text{resp. } <) p_h$ if $p_h > (\text{resp. } <) 1/2$. To see the rationale behind this effect, consider, for example, $p_h < 1/2$, that is, the low value is associated with a higher initial belief than the high value. The knowledge that the two products share a common value strengthens consumers' belief about either product being of low value. This is because the values are independently distributed and the two values coinciding with each other is more likely to occur when either is of low value than when either is of high value, that is, $\Pr[u_1 = u_2 | u_i = u_l] = p_l > \Pr[u_1 = u_2 | u_i = u_h] = p_h$.

As an implication of the reinforcement-by-homogeneity effect, if consumers (i) have little faith about the high value ex ante and (ii) are induced (e.g., by sales information) to believe that the products are very likely of same value, they become even more reluctant to search a product than without the sales information. Here, condition (ii) is satisfied when the sales difference is sufficiently small as exemplified by Lemma 5. We elaborate on the effect of sales difference by decomposing the posterior:

$$\begin{aligned} \pi_1^v(x) &= \Pr[u_{i^*} = u_h | X_{i^*} = x] \\ &= \Pr[u_{i^*} = u_h | u_1 \neq u_2, X_{i^*} = x] \Pr[u_1 \neq u_2 | X_{i^*} = x] \\ &\quad + \Pr[u_{i^*} = u_h | u_1 = u_2, X_{i^*} = x] \Pr[u_1 = u_2 | X_{i^*} = x]. \end{aligned}$$

Lemma 6 (Impact of First Period Sales Difference on Second Period Beliefs). *For $x \geq n_1/2$,*

- i. *Both $\Pr[u_r = u_h | u_1 \neq u_2, X_r = x]$ and $\Pr[u_1 \neq u_2 | X_r = x]$ increase in x .*
- ii. *$\pi_1^v(x)$ increases in x and $\pi_2^v(x)$ decreases in x for $x \geq n_1/2$.*

Lemma 6(i) reveals two primary drivers for the impact of sales volume information on consumer beliefs. First, the first half of part (i) proves that, conditional on the products having different values, a greater sales difference (i.e., higher bestseller sales x) renders the second period consumers more optimistic about discovering a high value in the bestseller. This is akin to the effect of sales ranking in Proposition 3. The underlying rationale is that it becomes increasingly difficult for a low-value product to achieve the corresponding sales advantage over a high-value one as the bestseller's sales x grows. Furthermore, by the second half of part (i), sales volume differs from sales ranking by impacting consumers' belief about whether product values are different from each other. Specifically, the greater the sales difference, the stronger the belief that the consumers hold about the two products being of different values. This is because a greater sales difference is increasingly more likely to arise when the products have different values than when they share the same value. Specifically, the greater the sales difference, the more likely that the sales difference is due to some early consumers' second searches and selective purchases (rather than purely random purchases) and, thus, the higher the chance that it reflects a distinction in the two products' values.

Synthesizing the two effects in part (i), Lemma 6(ii) proves that, under the volume information, the second period consumers hold a stronger belief about the bestseller's high value when the first period sales difference is greater. Likewise, conditional on the bestseller being of low value, a greater sales difference lowers consumers' expectation about the value revealed through a second search. The rationale is similar to that for Propositions 3 and 4(i): it is less likely for a low-value product to outsell a high-value product than it does another low-value product; upon observing a wider sales gap, consumers assign an even lower probability for the former relative to the latter.

4.3.3. Ranking Information vs. Volume Information.

Now, we compare consumers' beliefs under two types of sales information: ranking and volume. Our findings so far imply that, from the consumers' perspective, sales volume is more informative than sales ranking because observing the difference in sales volume allows them to update their beliefs about whether the products' values are identical and also to

refine their estimations about the probability that the bestseller is of high value. As noted by Yu et al. (2016), this higher informativeness can be captured by second order stochastic dominance of the posteriors. Lemma 7 follows.

Lemma 7 (Second-Period Posteriors: Ranking versus Volume). *$\pi_j^v(X_r)$ is a mean-preserving spread of π_j^r for $j = 1, 2$.*

The rationale behind Lemma 7 is simple: whereas both types of information allow the consumers to discern the bestseller, volume information is more refined and detailed than ranking information, which obscures and, thus, can correspond to any possible realization of sales differences. Consequently, the beliefs under ranking information are the expectation of the corresponding beliefs under volume information, that is, $\pi_j^r = \mathbb{E}[\pi_j^v(X_r)]$. By Lemmas 6 and 7, we have the following result.

Proposition 5 (First Search Beliefs: Ranking versus Volume). *There exists a threshold, $\bar{x}$, such that $\pi_1^v(x) \geq \pi_1^r$ if $x \geq \bar{x}$ and $\pi_1^v(x) < \pi_1^r$ if $x < \bar{x}$.*

Proposition 5 indicates that, when the sales difference is small, consumers are less confident about the bestseller being of high value under volume information than they are under ranking information. This result has direct implications on platform sales, on which we elaborate in the next section.

5. Impact of Sales Information on Platform Sales and Consumer Surplus

So far, we have focused on how sales information enables and influences social learning. These effects have direct implications for platform sales and consumer payoffs (e.g., average value of purchased product, expected search cost incurred before purchasing, and aggregate surplus) as examined next.

5.1. Platform Sales

By Lemma 3, we denote the expected second period sales by $S_t := \mathbb{E}[n_2 F(u_h \pi_1^t + u_l(1 - \pi_1^t))]$ for $t \in \{\phi, r, v\}$, where the expectation is taken over π_1^t (recall that both π_1^ϕ and π_1^r are constants, whereas π_1^v is a function of the bestseller's sales, X_r , which is a random variable ex ante). The comparative results of the beliefs in Propositions 3 and 4 imply their counterparts for the sales as follows.

Proposition 6 (Impact of Sales Information on Second Period Sales). *Compared with no information,*

- i. *Ranking information increases the expected second period sales, that is, $S_r \geq S_\phi$. In particular, there exist problem instances in which $S_\phi/S_r \leq \epsilon$ for any given $\epsilon \in (0, 1)$.*
- ii. *Volume information reduces the second period sales if both $p_h < 1/2$ and the first period sales difference is small*

(specifically, when the bestseller's sales volume X_{i^*} is below a threshold $\tilde{x}$) and increases the sales otherwise. Furthermore, volume information may lead to a lower expected second period sales; that is, there exist problem instances in which $S_v < S_\phi$.

By Proposition 6, the platform always finds it advantageous to provide some sales information to the consumers, and yet, the more refined information may not necessarily be its preferred choice. More specifically, sales ranking information always benefits the platform and may improve its expected sales by a ratio arbitrarily high, whereas providing information of higher precision about sales volume may backfire. In particular, the expected value of volume information can be negative to the platform. As aforementioned, this is because ranking information enhances consumers' confidence about discovering high value in the bestseller, whereas volume information may dampen it.

This result bears some interesting resemblance with Yu et al. (2016), who focus on a single-product setting and show that a seller always gains from noisy consumer reviews and yet may get worse off by accurate review information. Whereas both results are derived under social learning frameworks, the underlying drivers are drastically different. Specifically, the value of review information in Yu et al. (2016) originates from the seller's option value of dynamically adjusting prices according to the reviews and the consumers' option value of strategically postponing their purchasing decision until the reviews are available. When review information is noisy, consumers are reluctant to wait and the seller benefits from information-induced price adjustment. Accurate reviews, however, reinforce consumers' waiting incentive, and this is disadvantageous to profits and, thus, can hurt the seller.

In contrast, sales information in our two-product context draws its value from two possible sources. First, regardless of its specific type (ranking or volume), sales information converts consumers' random product search to directional search starting from the product with a higher expected value, which thus strengthens consumers' motivation to explore the products and benefits the platform. This effect is specified in the first line of (2). Second, sales volume information enables social learning of the products' joint value distribution, and this may, however, render the consumers less willing to search and, thus, lower the overall product sales. As noted earlier, this effect is unique to sales volume information and as specified in the second line of (2). Consequently, sharing sales ranking with consumers is always beneficial to the platform, whereas releasing information about sales volume

may backfire:

$$\begin{aligned} \pi_1^t - p_h &= \Pr[u_{i^*} = u_h | X_{i^*}] - p_h \\ &= \Pr[u_{i^*} = u_h | X_{i^*}] - \frac{1}{2} (\Pr[u_{i^*} = u_h | X_{i^*}] + \Pr[u_{-i^*} = u_h | X_{i^*}]) \\ &\quad \underbrace{\qquad\qquad\qquad}_{\text{Gain of a directional first search over a random first search,}} \\ &\quad \underbrace{\qquad\qquad\qquad}_{\text{"} \geq 0 \text{" for both ranking and volume, "} = 0 \text{" for volume when } X_{i^*} = \frac{n_1}{2}} \\ &+ \frac{1}{2} (\Pr[u_{i^*} = u_h | X_{i^*}] + \Pr[u_{-i^*} = u_h | X_{i^*}]) - p_h \\ &\quad \underbrace{\qquad\qquad\qquad}_{\text{Change in the probability of a randomly-selected product being of high value,}} \\ &\quad \underbrace{\qquad\qquad\qquad}_{\text{"} = \frac{1}{2} (2\Pr[u_1 = u_2 = u_h | X_{i^*}] + \Pr[u_1 \neq u_2 | X_{i^*}]) - p_h \text{"}} \\ &\quad \underbrace{\qquad\qquad\qquad}_{\text{"} = 0 \text{" for ranking, "} >, <, > < \text{" for volume.}} \end{aligned} \quad (2)$$

Given that volume information may negatively impact product sales, the platform sometimes favors ranking information over volume information. Proposition 7 follows.

Proposition 7 (Second Period Sales: Ranking versus Volume). *Compared with ranking information,*

i. *Volume information reduces the second period sales when the first period sales difference is small (specifically, when the bestseller's sales volume X_{i^*} is below $\tilde{x}$) and increases the sales otherwise.*

ii. *Volume information may lead to lower expected second period sales. In particular, when the search cost distribution $F(\cdot)$ is convex, $S_r \leq S_v$, and when $F(\cdot)$ is concave, $S_r \geq S_v$. Furthermore, there exist problem instances in which $S_r/S_v \leq \epsilon$ for any given $\epsilon \in (0, 1)$.*

Proposition 7(i) is implied by Proposition 5, and its key driver is the fact that, given a small first period sales difference, the gain of a directional first search over a random first search, as defined in the first line of (2), is lower under volume information than ranking information. To see the rationale behind, note that, when the sales difference is small, the search direction implied by the first period sales is of little help to the consumers as it is very likely caused by first period consumers' random purchases instead of their deliberated choice of a high-value product after searching both products. Specifically, a small sales difference may imply that either the products almost certainly have identical values or their values have been verified by few early buyers or both. Under any of these situations, whether to first search the bestseller or the other product is of little importance to the second period consumers. This effect is most prominent when the sales difference is zero, that is, $x = n_1/2$: the products remain symmetric to the second period consumers upon observing identical first period sales volume, and thus, a directed search is as good as a random search. In contrast, sales ranking conceals sales volume difference and, thus, disallows the late consumers to draw any inference about whether the early consumers' purchases are randomized or selective. Conditional on some early consumers having

searched both products, ranking information always creates a positive gain of directional search by the simple fact that a higher value product (if any) is more likely to be the bestseller.

Part (ii) of Proposition 7 follows from the property of mean-preserving spread in Lemma 7 and presents a sufficient condition for the platform's preference, ex ante, for ranking information over volume information. It is interesting to note the role of search cost distribution in determining the preferred type of sales information for the platform. In particular, if the distribution is concave, providing bestseller rankings outperforms disclosing sales volumes. These findings suggest a plausible explanation for platforms' different practices of sales information provision (e.g., bestseller ranking on Amazon.com and sales volume on Taobao.com) as noted in the introduction. Specifically, as ranking information can induce higher expected sales than volume information, publicizing an ordered list of bestsellers while concealing detailed sales numbers can be a platform's optimal disclosure strategy.

Furthermore, Proposition 7(ii) also reveals that the percentage gap in the expected sales under volume information and that under ranking information can be arbitrarily large. This finding underscores the significance of evaluating and comparing across different information provision policies.

5.2. Consumers' Perspective: Product Utility, Search Cost, and Aggregate Surplus

Now, we switch to the consumers' viewpoint and examine the impact of sales information on average value of a purchased product, expected search cost incurred, and overall consumer surplus.

5.2.1. Average Value of a Purchased Product. Given $t \in \{\phi, r, v\}$, the expected product value acquired by a late arriving consumer with search cost s is $\mathbb{E}[(u_h - u_l)(\pi_1^t + (1 - \pi_1^t)\pi_2^t\mathbb{I}(s \leq \pi_2^t(u_h - u_l))) + u_l]\mathbb{I}(s \leq \pi_1^t u_h + (1 - \pi_1^t)u_l)$, where $\mathbb{I}(\cdot)$ is the indicator function and the expectation is taken over the posteriors. Proposition 8 reveals an interesting effect of (finer) sales information on the expected purchased value.

Proposition 8 (Impact of Sales Information on Expected Purchased Value). *Consider a second period consumer with search cost s :*

- Compared with the case in which no sales information is provided, ranking information provision reduces the consumer's expected purchased value if $\pi_2^t(u_h - u_l) < s \leq p_h(u_h - u_l)$.*
- Compared with the case in which ranking information is provided, volume information provision reduces the consumer's expected purchased value if $\pi_2^v(n_1)(u_h - u_l) < s \leq \pi_2^r(u_h - u_l)$.*

It is perhaps surprising to note that (finer) sales information does not always lead to an improvement in a consumer's expected purchased value. As in Proposition 8, because of the platform's disclosure of (better) sales information, consumers with intermediate search costs may, with a higher chance, purchase a low-value product. This is driven by a reduction in their second search probability. In particular, as we note, ranking information dampens consumers' belief about discovering a high value in the lower ranked product, and the belief is further lowered when the bestseller is found to be of low value and/or when a significant difference is revealed in the first period sales. Therefore, (better) sales information may induce these consumers to forsake the opportunity of searching the lower ranked product, which may, however, turn out to be of high value in the unfortunate event that many early consumers have randomly picked the low-value product in their sole search such that the low-value product outsells the high-value one in the first period.

Proposition 8 indicates that, because of social learning, consumers may disregard the chance to seek more private information and, consequently, end up with lower purchased value on expectation. This result is similar to the well-known phenomenon of information cascade (Banerjee 1992, Bikhchandani et al. 1992), in which consumers ignore their personal knowledge and follow others' choices of suboptimal options. The key driver for information cascade is the inherent noise in the early buyers' private signal, which misleads the actions of their own and also of the late arriving consumers. In contrast, consumers' private information in our model is perfect, but because of the search cost, some early consumers cannot afford to acquire the information about all the products. For these consumers, which product to search first is essential as it is also the one that they will purchase, and yet, this first search product choice can only be made through random draws. These buyers' incomplete product information and random first search choices jointly lead to the noise in the first period sales information, which may misguide some second period consumers to overlook a high-value product.

5.2.2. Expected Search Cost Incurred Before Purchasing. As alluded to previously, consumers' product search represents their private information acquisition. Thus, one may naturally wonder: are consumers less likely to acquire private information when (finer) public information is available? Consider again a consumer with search cost s . By the prior analysis, the consumer's probability of performing the first search and that of the second search are $\mathbb{E}[\Pr(s \leq \pi_1^t u_h + (1 - \pi_1^t)u_l)]$ and $\mathbb{E}[\Pr(s \leq \pi_2^t(u_h - u_l))]$, respectively, where the expectations are taken over the posteriors. The

effects of (more) sales information on the search probabilities are summarized below.

Proposition 9 (Impact of Sales Information on Search Probabilities). *Consider a second period consumer with search cost s :*

- i. *Compared with the case in which no sales information is provided, ranking information provision increases the first search probability and decreases the second search probability.*
- ii. *Compared with the case in which ranking information is provided, volume information provision decreases both search probabilities if $\pi_2^v(n_1)(u_h - u_l) < s \leq \pi_2^r(u_h - u_l)$, and increases both probabilities if $\pi_1^r u_h + (1 - \pi_1^r)u_l < s \leq \pi_1^v(n_1)u_h + (1 - \pi_1^v(n_1))u_l$.*

Whereas one may expect that (finer) public information renders consumers less likely to spend costly efforts verifying product values themselves, Proposition 9 discovers that the opposite may occur. This is because sales information can enhance consumers' confidence about discovering a high-value product through their searches and, thus, increase their willingness to explore more products. Specifically, this is achieved by guiding consumers to the product with higher expected value in first search or/and informing them of the sales difference to strengthen their belief about finding high value in second search. The latter effect is especially pronounced for consumers with high search costs, who would not explore any product if only sales ranking is provided and yet may search under volume information. In other cases, sales information may render consumers more pessimistic about product values and, thus, disincentivize their searches: consumers who have relatively low search costs and would search twice under ranking information may be reluctant to search both products if sales volume is disclosed.

It is noteworthy that, as implied by Proposition 9, public learning can be complementary to private learning as public sales information may promote private product search.

5.2.3. Aggregate Consumer Surplus. Now, we synthesize the analysis of purchased value and search cost to examine consumers' expected surplus. A consumer's expected surplus equals to the consumer's expected purchased value net of the consumer's expected search cost: specifically, the expected surplus of a consumer with search cost s is $\mathbb{E}[\max[0, \pi_1^t u_h + (1 - \pi_1^t)u_l, \pi_2^t u_h + (1 - \pi_2^t)u_l - s] - s]$, where the two maximum operators correspond to the consumer's decisions on the two searches. Aggregate consumer surplus can be obtained by integrating individual surplus over all possible search costs. Interestingly, whereas sales information may lead to either lower purchased value or higher search cost, its overall impact on (aggregate) consumer surplus is always positive, and its value to

consumers increases as the information becomes finer. Proposition 10 follows.

Proposition 10. *A consumer's expected surplus is higher under either ranking or volume information than that under no information. Furthermore, it is higher under volume information than that under ranking information.*

Proposition 10 confirms that, as sales information enables social learning, it allows consumers to make better informed decisions and, thus, is beneficial to them. That is, whereas sales information may negatively impact on either purchased value or search cost, it never does so on both aspects, and its negative effect on one aspect, if any, is dominated by its positive effect on the other. This is because, as rational decision makers, consumers adjust their decision in response to (better) sales information if and only if doing so improves their expected surplus. Therefore, (better) sales information never hurts the consumers. Interestingly, this result echoes the empirical findings in Fu (2020) that, when products are randomly ranked and no sales information is available in the initial period (as in our model), disclosing initial period sales information to subsequent consumers enhances their welfare.

It is noteworthy that Proposition 10 is in sharp contrast to our findings regarding the platform, which may prefer disclosing sales ranking rather than sales volume. Thus, the platform may withhold information at the cost of consumers. On the other hand, to achieve long-term benefits, some platforms may adopt a consumer-centric objective of maximizing consumer welfare (Derakhshan et al. 2022). Proposition 10, thus, suggests that such platforms disclose sales volume in their bestseller lists.

6. Optimal Timing of Sales Information Provision

We proceed to examine the platform's optimal timing of sales information provision, that is, the optimal decision on n_0 , the number of consumers arriving in the first period. As the total number of consumers in the two periods is fixed, the decision of n_0 reflects a typical trade-off between information exploration and exploitation: as n_0 increases, the sales information becomes more informative about the products' values because more consumers search both products in the first period, and yet fewer consumers arrive in the second period to learn from the information. It remains unclear how the provision timing impacts on the platform's preference between ranking and volume information, which we explore next.

We first analyze the effect of increasing n_0 on the two-period total expected sales under either ranking or volume information based on which we then examine how the preferred information type varies by n_0 .

Because analytical characterization of the effect of n_0 is very difficult, we resort to numerical study. When choosing the search cost distribution, we refer to the empirical estimation in Moraga-González et al. (2013), which finds that “the search cost density is essentially bimodal such that a large fraction of consumers searches very little, whereas a smaller fraction searches a relatively large number of stores.” (p. 1205) In particular, the authors note that the group of consumers searching a lot have search costs close to zero, and apart from this consumer group, the search cost density has a unimodal structure in which the density first increases and then decreases. Accordingly, we assume in our numerical study that the search cost density is bimodal: specifically, $F(x) = [\alpha + (1 - \alpha)\Phi((x - \mu)/\sigma)]\mathbb{I}(x \geq 0)$, where $\Phi(\cdot)$ is the cumulative distribution function for the standard normal distribution and $\mathbb{I}(\cdot)$ is the indicator function. That is, the mass at $x = 0$ represents the small proportion of consumers who have small search costs and the scaled normal distribution represents the remaining consumers. We further set $\alpha = 0.08$ and $\mu = 4.5$ as approximated according to figure 1 in Moraga-González et al. (2013).

For a fixed n , Table 1 presents the effects of n_0 on the total expected sales of the two products in the two periods. Under either ranking or volume information, as n_0 increases, the two-period expected sales first grows and then declines, reflecting the aforementioned interplay between information exploration and exploitation. Interestingly, whereas the first period sales increases proportionally in n_0 , the second period sales under either information may initially increase and then drop in n_0 because of the joint effect of two opposing forces: on the one hand, as n_0 increases, sales information becomes more informative, which enables the platform to motivate more searches (and,

thus, purchases) from the second period buyers. On the other hand, an increase in n_0 reduces the number of consumers in the second period, naturally limiting the platform’s sales potential from exploiting the information. Thus, when n_0 is close to either zero or n , sales information creates little value for the platform because of inadequate exploration or exploitation. For intermediate values of n_0 , the increment in the two-period expected sales because of information differs by the type of information. As illustrated in Table 1 (e.g., $\mu = 6.5$ and $n_0 = 160$), the sales improvement achieved by bestseller information can be as high as 110%, and the percentage difference in the two-period sales under ranking and volume information can be more than 35%.

Whereas the platform-preferred information type is independent of n_0 in Table 1, we note in Table 2 an interesting pattern regarding how the preferred information type varies by n_0 : under low n_0 , volume information leads to higher sales than ranking information, whereas the opposite applies when n_0 is high. We observe this pattern whenever the preferred information type varies by n_0 and conjecture that it is due to the shape of the search cost density as explained below.

First note that it suffices to consider only the second period when comparing the expected two-period sales under different sales information as the information does not affect the first period. Recall from Section 5.1 that the expected second period sales under volume information $S_v = \mathbb{E}_{X_r} [n_2 F(u_h \pi_1^v(X_r) + u_l(1 - \pi_1^v(X_r)))]$, where $\pi_1^v(X_r)$ is a mean-preserving spread of π_1^r on the support $[\pi_1^v(n_1/2), \pi_1^v(n_1)]$. Let $EU(\pi) := u_h \pi + u_l(1 - \pi)$ denote a product’s expected value when the probability of high value is π . By Jensen’s inequality, if $F(\cdot)$ is convex (concave) on the interval $\Omega := [EU(\pi_1^v(n_1/2)), EU(\pi_1^v(n_1))]$, the expected second period sales under

Table 1. Two-Product Sales as Functions of n_0 and Information Type t

n_0	$\mu = 6.5$							$\mu = 3.5$						
	First period	Second period			Two periods			First period	Second period			Two periods		
		$t = v$	$t = r$	$t = \phi$	$t = v$	$t = r$	$t = \phi$		$t = v$	$t = r$	$t = \phi$	$t = v$	$t = r$	$t = \phi$
10	1.04	103.4	103.4	103.4	104.5	104.5	104.5	5.64	642.8	644.3	558.8	648.5	650.0	564.5
60	6.27	186.1	159.9	98.2	192.3	166.2	104.5	33.9	700.4	726.5	530.6	734.3	760.4	564.5
110	11.5	204.1	151.4	93.0	215.6	162.9	104.5	62.1	651.6	693.0	502.4	713.7	755.1	564.5
160	16.7	203.3	142.9	87.8	220.1	159.6	104.5	90.3	607.7	654.5	474.1	698.0	744.8	564.5
260	27.2	183.9	125.9	77.3	211.0	153.1	104.5	146.8	526.9	576.7	417.7	673.6	723.4	564.5
360	37.6	159.9	108.9	66.9	197.5	146.5	104.5	203.2	453.3	498.8	361.3	656.5	702.0	564.5
460	48.1	135.1	91.9	56.4	183.1	139.9	104.5	259.7	381.6	420.8	304.8	641.2	680.5	564.5
560	58.5	110.1	74.9	46.0	168.6	133.4	104.5	316.1	310.7	342.9	248.4	626.8	659.0	564.5
660	69.0	85.1	57.8	35.5	154.0	126.8	104.5	372.5	240.0	265.0	191.9	612.5	637.5	564.5
760	79.4	60.0	40.8	25.1	139.4	120.2	104.5	429.0	169.4	187.0	135.5	598.4	616.0	564.5
860	89.8	35.0	23.8	14.6	124.9	113.7	104.5	485.4	98.8	109.1	79.0	584.2	594.5	564.5
960	100.3	10.0	6.8	4.18	110.3	107.1	104.5	541.9	28.2	31.2	22.6	570.1	573.0	564.5

Note. $n = 1,000$, $p_h = 0.4$, $\alpha = 0.08$, $u_h = 6$, $u_l = 2$, and $\sigma = 1.5$, where the boldface numbers correspond to the platform-preferred information type for given n_0 .

Table 2. Impact of n_0 on Total Expected Sales in Two Periods

n_0	$u_h = 6, u_l = 2$				$u_h = 6.5, u_l = 2.5$			
	Ranking	Volume	No information	Opt. information	Ranking	Volume	No information	Opt. information
0	33.23	33.23	33.23	Tie	44.33	44.33	44.33	Tie
10	42.79	42.91	33.23	Volume	52.15	52.16	44.33	Volume
20	44.01	44.29	33.23	Volume	54.51	54.41	44.33	Ranking
40	44.04	44.10	33.23	Volume	54.56	54.31	44.33	Ranking
60	41.26	41.25	33.23	Ranking	51.86	51.57	44.33	Ranking
80	37.38	37.36	33.23	Ranking	48.40	48.19	44.33	Ranking
100	33.23	33.23	33.23	Tie	44.33	44.33	44.33	Tie

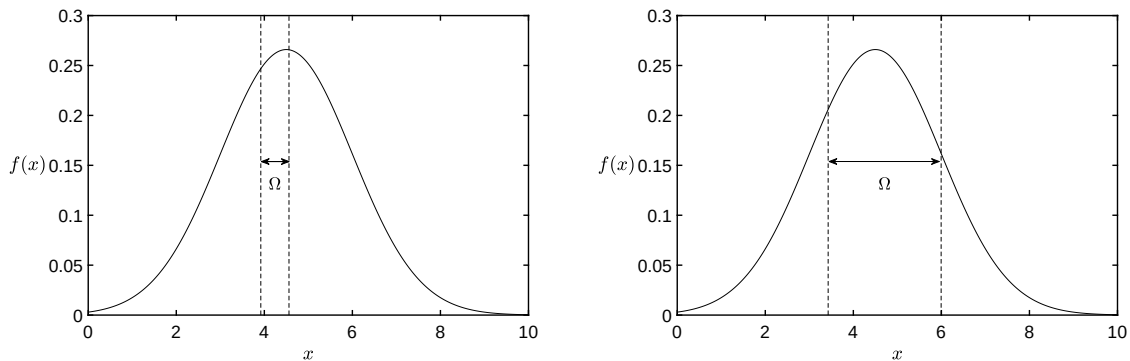
Note. $\mu = 4.5, \sigma = 1.5, n = 100, p_h = 0.4, \alpha = 0.08$.

volume information is higher (lower) than that under ranking information, that is, $S_v \geq (\text{resp. } \leq) S_r$. Besides, we verified, both numerically and analytically under some approximation, that the upper bound of the interval Ω , $EU(\pi_1^v(n_1))$, increases in n_0 . Thus, it is possible that, when n_0 is small, the interval Ω lies (almost) entirely within the interval $(0, \mu)$, where the search cost density increases, and thus, $F(\cdot)$ is convex, resulting in a higher expected sales under volume information than that under ranking information (see Proposition 7(ii) and the left panel of Figure 6). As n_0 increases, the upper bound of the interval Ω increases, and thus, a larger portion of the interval may fall into the region above μ , where the search cost density decreases and $F(\cdot)$ is concave (see the right panel of Figure 6). Hence, we observe that ranking information is increasingly likely to outperform volume information at a larger n_0 . For a similar reason, a stochastically higher product value (e.g., when both u_h and u_l are increased by a same amount) shifts the interval Ω to the right and, thus, tends to render ranking information more often preferred by the platform to volume information as we observe by comparing the left and right panels of Table 2. Likewise, when μ increases such that the entire density curve shifts to the right, volume information may

more likely emerge as the preferred one as a greater portion of the interval Ω may fall within $(0, \mu)$. This is observed in Table 1, in which the preferred information switches from ranking to volume as μ increases from 3.5 to 6.5.

We obtain similar results when both the timing and type of sales information provision are endogenous. Table 3 reports the optimal n_0 , denoted by n_0^* , and the corresponding preferred information type for different values of total population size, n . Observe from the table that the larger the consumer population, the later the sales information should be released; that is, n_0^* increases in n . Intuitively, for a given n_0 , a larger population, n , implies more consumers arriving in the second period, which allows the platform to gain more from the sales information. That is, a growth in the total market mitigates the negative impact on information exploitation by a higher n_0 , and induces the platform to expand the exploration stage. We prove this result in Section SK for some special case and under some approximation. Furthermore, similar to that in Table 2, ranking information is more likely to be preferred by the platform under either a larger consumer population or a stochastically higher product value.

Figure 6. Search Cost Density, Denoted by $f(x)$, and the Interval Ω When $n_0 = 10$ (Left Panel) and $n_0 = 100$ (Right Panel): $p_h = 0.4, u_h = 6, u_l = 2, \mu = 4.5, \sigma = 1.5, n = 100$



Note. The mass point at $x = 0$ is omitted in the figure.

Table 3. Impact of n on Total Expected Sales in Two Periods

n	$u_h = 6, u_l = 2$						$u_h = 6.5, u_l = 2.5$					
	Ranking		Volume		Opt. information		Ranking		Volume		Opt. information	
	n_0^*	Sales	n_0^*	Sales	n_0^*	Type	n_0^*	Sales	n_0^*	Sales	n_0^*	Type
20	10	7.71	10	7.72	10	Volume	10	9.73	10	9.74	10	Volume
50	10	20.87	10	20.92	10	Volume	20	25.98	20	25.95	20	Ranking
100	29	45.38	29	45.46	29	Volume	29	55.18	29	54.98	29	Ranking
1,000	96	527.68	77	525.63	96	Ranking	115	636.99	96	623.43	115	Ranking

Note. $\mu = 4.5, \sigma = 1.5, p_h = 0.4, \alpha = 0.08$.

7. Extensions

7.1. Strategic Consumer Waiting for Bestseller Information

In practice, some sophisticated consumers may strategically postpone their decisions until after bestseller information is available. We extend the model in Section SL to incorporate such behavior. Specifically, we consider the situation in which all of the n consumers arrive up front and decide simultaneously and independently on whether to make the search and purchasing decisions in the first period or postpone the decisions to the second period. Postponement is costly as future utility is discounted (at a rate δ) and yet can be beneficial in anticipation of the sales information available in the second period.

Formally, for a consumer with search cost s , the expected utility without decision postponement is

$$U_1(s) = \max[0, p_h u_h + p_l \max(u_l, p_h u_h + (1 - p_h) u_l - s) - s],$$

and under information type $t \in \{\phi, r, v\}$, the expected utility by postponing decisions to period 2 is

$$U_2^t(s; \pi_1^t(\cdot), \pi_2^t(\cdot)) = \mathbb{E}[\delta \max[0, \pi_1^t u_h + (1 - \pi_1^t) \max(u_l, \pi_2^t u_h + (1 - \pi_2^t) u_l - s) - s]],$$

where the posteriors $\pi_1^t(\cdot)$ and $\pi_2^t(\cdot)$ are endogenously determined by all the consumers' choice on postponement. Decision postponement is optimal if and only if $U_1(s) \leq U_2^t(s; \pi_1^t(\cdot), \pi_2^t(\cdot))$.

We first show that, for given posteriors, consumers with lower search costs are less likely to postpone decisions as they expect a higher utility from the decisions and, thus, a greater loss of utility (because of discounting) if they wait. Based on this result, we prove that a threshold-type equilibrium exists among consumers in which those with search cost less than a threshold, $\hat{s}^t$, make their searching and purchasing decisions in the first period and others postpone.

As consumers become increasingly strategic (i.e., θ increases), the threshold $\hat{s}^t$ declines and fewer consumers purchase early. This may, however, enhance the informativeness of the first period sales. (In particular, we show that $\mathbb{E}[\pi_1^t(\cdot)]$ increases as the threshold $\hat{s}^t$ decreases. That is, the probability of the bestseller

being of high value is higher, on expectation, as fewer consumers purchase early). This is because the consumers who self-select to purchase early are those with low search costs, and thus, their purchasing decision are mostly informed (i.e., after two product searches). Hence, the more patient the consumers are, the more consumers with high search cost postpone decisions and the higher the proportion of informed early purchases. This renders sales information less noisy, and this can substantiate the willingness to search of the consumers who procrastinate, thus promoting the second period sales and benefiting the platform. Whereas strategic consumer behavior is often detrimental to firms, it can be advantageous to the platform under social learning through bestseller information.

Furthermore, compared with ranking information, volume information induces more consumers to postpone purchases as they expect a higher second period surplus under sales information of a higher granularity, which resonates with our finding that consumers benefit more from finer sales information. This result also implies that the platform can provide bestseller information of a lower granularity if it hopes to dampen consumers' waiting incentives. In addition, similar to that in the base model, ranking information may outperform volume information and induce higher total sales over the periods.

7.2. Personalized Information Provision

Thus far, we have assumed that sales information provision is public. That is, consumers access to the same (type of) information if the platform offers it. This is in line with platforms' prevailing practices. In Section SM, we explore the situation when the platform is aware of each consumer's search cost and customizes the type of sales information provision (ranking or volume) accordingly. We find that the optimal personalized provision strategy is characterized by a cutoff on a consumer's search cost.

Proposition 11 (Personalized Information Provision). *It is optimal for the platform to provide ranking (volume) information to a consumer if the consumer's search cost, s , is less than or equal to (greater than) $\pi_1^r u_h + (1 - \pi_1^r) u_l$.*

The rationale behind Proposition 11 is as follows. As the platform aims to maximize sales, it personalizes information provision to maximize the probability of purchasing for each consumer. Recall that consumers with search cost less than or equal to $\pi_1^r u_h + (1 - \pi_1^r) u_l$ purchase for sure under ranking information and yet may not purchase under volume information because volume information may dampen their belief about the bestseller being of high value (i.e., π_1^r may be lower than π_1^v). Thus, it is optimal for the platform to only disclose to those consumers sales ranking. For consumers with search cost greater than $\pi_1^r u_h + (1 - \pi_1^r) u_l$, they never purchase under ranking information but may be induced to buy if they are informed that the sales difference between the products is significant ($\pi_1^v > \pi_1^r$ in such cases). Consequently, it is optimal to provide these consumers with volume information.

In terms of welfare implication, the platform never gets worse off by customizing its information provision as it represents a higher level of operational flexibility. Nevertheless, information personalization may be detrimental to the consumers. Specifically, as consumer surplus increases in the information level, when information personalization results in less information offered to some consumers, it reduces aggregate consumer surplus. That is, a platform may, in its own interest, individualize its information provision, which may end up hurting the consumers as a whole.

7.3. Other Extensions

7.3.1. Generalized Product Settings. In Section SB, we show that our main results remain valid under three products. Specifically, we prove that, compared with no information, ranking information enhances the second period expected sales, whereas volume information may reduce it. We also observe that ranking information's sales-enhancing effect may be more pronounced as the number of products increases: given ranking information, consumers may become more confident about the bestseller's value and more willing to perform a first search under three products than under two products.

Furthermore, we analyze in Section SC a generalized setting in which the two products are ex ante asymmetric with different prior probabilities of high value. Besides confirming the robustness of the main results, we find that product asymmetry may reduce the variability of the first period sales as the uninformed consumers (i.e., those willing to search only once) always purchase the more promising product. This effect can render sales ranking less informative as, in some cases, the more promising product always has higher sales. On the other hand, this effect results in purchases by the informed consumers (i.e., those willing to search twice) deducible from the realized

sales volume and, consequently, product asymmetry can enhance the informativeness of sales volume.

7.3.2. Additional Consumer Heterogeneity. The base model assumes that all the consumers' reservation utility is zero. In Section SN, we relax this assumption to consider the situation in which the reservation utility is heterogeneous across consumers and may be higher than u_l .⁸ Under consumer heterogeneity in both search cost and reservation utility, we analyze two scenarios: (i) when search is required for purchase and (ii) when consumers may purchase without search. In both scenarios, we confirm our main insights and also find that sales volume becomes more informative of product values under heterogeneous reservation utility because consumers with high reservation utility only purchase a high-value product, and thus, the total first period sales allows inference about whether both products have low value. Furthermore, for scenario (ii), we note that the search probabilities may increase owing to availability of sales information, which strengthens the finding regarding complementarity between public learning and private learning as search is now for the sole purpose of information acquisition.

7.3.3. Platform's Real-Time Information Provision. To evaluate the impact of more frequent information provision, we consider in Section SP the setting in which sales information is updated in real time and after each consumer's purchasing decision. Compared with the base model, real-time information can allow consumers to acquire better knowledge about product values. This can be attributed to the fact that real-time information provision allows the later arrivals to leverage both public and private learning conducted by their predecessors. Nevertheless, more frequent information provision may be undesirable to the platform as better informed consumers may lead to lower product sales.

7.3.4. Learning-Induced Pricing Competition. Whereas pricing is not our focus, Section SQ presents an extension in which two sellers, either offering one of the products, simultaneously and independently choose a second period price of their own product upon observing the bestseller information released by the platform. We find that the sellers' flexibility of adapting prices to information may render the platform increasingly likely to prefer sales volume information over ranking information and that the platform may postpone its bestseller information provision because of pricing flexibility.

8. Conclusion

Motivated by major platforms' diverse practices in bestseller information provision, this paper examines

consumers' learning, search, and purchasing behavior when choosing between two *ex ante* homogenous products for which a platform strategically decides on disclosure of historical sales information. It embeds consumers' sequential search in a social learning framework and demonstrates its key role in the generation of bestseller information. It also reveals how bestseller information of varying granularity impacts late arriving consumers' sequential search through social learning. We show that the interaction between bestseller information and consumer search has important implications to the platform's optimal choices on the granularity and timing of bestseller information provision.

Specifically, a bestseller list can be indicative of product value difference because of the selective purchases by those early arriving consumers who research both products, but the signal it conveys is imperfect because of the random purchases by those who search only once. Compared with the situation in which either no sales information or only sales ranking is revealed, disclosing numbers of units sold allows more accurate social inference about how likely some of the early consumers' purchases are selective and, thus, enables more effective detection of any difference in product values. In contrast, sales ranking disallows social learning of whether product values differ from each other. The gap that sales ranking creates in expected product values is merely by the fact that, conditional on different values, a higher value product is more likely to achieve a top-selling status.

We prove that, owing to these differences, late arriving consumers always prefer sales volume information to ranking information as insights derived from the former lead to more informative search and purchasing decisions. Nevertheless, the same preference may not apply to the platform. In particular, sales ranking always benefits the platform by promoting consumers' search of the higher ranked product and enhancing overall product sales, whereas sales volume can dampen consumers' incentive to research or buy either product because of the reinforcement-by-homogeneity effect and be detrimental to the platform. Thus, consistent with the prevalence and various granularity levels of bestseller lists on e-commerce shopping platforms, we show that a platform is better off revealing at least sales rankings of its products but may or may not find it advantageous to disclose detailed sales volume. Important determinants of the latter include distributions of search cost and product value and sizes of consumer cohorts arriving over time. We find that convex (concave) search cost distributions ensure the platform's preference of volume (ranking) information provision, and under a particular search cost distribution empirically derived by a prior study in the search literature, stochastically higher product values and/or a larger size of either

the early arriving cohort or total consumer population can tilt the platform's preference toward disclosing ranking information only. These results also suggest that revealing sales volume can be particularly effective for promoting sales of products with relatively low perceived values and/or limited sales potential (e.g., targeted for niche markets).

In addition, the impact of sales information on consumers may not always be intuitive. Specifically, for some consumers, information (of higher granularity) may result in lower purchased value or higher search cost on expectation. The former arises because of a reduction in consumers' second search probability owing to sales information, and the latter implies that public learning may not substitute private learning. Besides, personalized information provision may not be a blessing to all consumers.

We confirm our main findings in several model extensions on products, consumers, and information provision strategies. Interestingly, strategic consumer waiting for bestseller information can be advantageous to the platform, whereas higher frequency information updating may backfire.

Our results lead to a conjecture that the shopping platforms' different practices in bestseller information provision could possibly reflect differences in consumers' search cost distribution and quality distribution of the products that consumers consider purchasing in the United States and in China. As detailed in Section SA.2, these differences are evidenced by distinctive product search and shopping behavior of American and Chinese consumers, which may be attributed to various cultural and market factors. Our results, thus, present interesting and testable hypotheses for empirical studies in these markets.

As alluded to earlier, our model abstracts from various other factors that may impact platforms' decision on sales information disclosure, rendering our insights about the decision largely normative. One other factor can be sellers' preference. For example, Apple initially provided download counts of apps in its App Store but later removed the information because of developers' objections.⁹ In addition, platforms may withhold sales figures as they are considered proprietary business information and, thus, highly sensitive commercially. Platforms may be stipulated, by regulations like the Defend Trade Secrets Act of 2016 in the United States, to protect sales information for sellers' best interests. They may also be under an obligation to safeguard sales information because of no-disclosure agreements with sellers.

The bestseller information reflects products' cumulative sales performance over an extended period of time. It helps consumers gain insights into products that deliver high values inherently and, thus, are successful consistently. In practice, platforms may provide other types of sales-related information. For example,

the Amazon Movers and Shakers list presents products with the most significant improvements in their sales rankings over the past 24 hours.¹⁰ It is useful for identifying trending products with low cumulative sales but fast rising popularity. As some short-term sales growth can be attributed to fleeting fads, inferring product values from the Movers and Shakers list is more complex than that from a bestseller list and poses an interesting question for future research.

Acknowledgments

The authors thank the department editor, an anonymous associate editor, and two anonymous referees for constructive comments that led to significant improvements of the paper.

Endnotes

¹ See www.amazon.com/Best-Sellers/zgbs.

² We consider generalized product settings such as three products, asymmetric products ex ante, and product value more than two levels and show that our key results are robust in these general settings (see Sections SB, SC, and SD).

³ In Section SE, we endogenize consumer arrivals by incorporating consumers' cost of visiting the platform.

⁴ We confirm in Section SF the main results when a consumer's search cost is independently drawn from the distribution $F(\cdot)$.

⁵ We discuss in Section SG the platform's alternative objectives and benchmark in Section SH against the first best scenario involving a welfare-maximizing principal.

⁶ Here, product ranking is solely determined by sales and represents a subset of the information that sales volume provides. Thus, it differs from other product rankings in practice, for example, those calculated by platform algorithms incorporating various factors such as consumer ratings (e.g., the star systems on Amazon or Uber). Considering ranking that is not a one-to-one mapping from sales requires modeling of nonsales factors and is beyond the paper scope. Whereas we primarily focus on sales ranking and volume, we show in Section SI that our key insights hold under a general class of sales signals, which is based on sales interval partitioning and subsumes ranking and volume information as special cases.

⁷ Our main results are robust if the bestseller information is released when some sales milestone is achieved, as in Section SJ.

⁸ We also consider in Section SO consumer heterogeneity in their individual valuation of the products and show that the model is equivalent to either the one with heterogenous reservation utility or the base model.

⁹ See www.wsj.com/articles/the-mobile-industrys-never-seen-any-thing-like-this-an-interview-with-steve-jobs-at-the-app-stores-launch-1532527201.

¹⁰ See www.amazon.com/gp/movers-and-shakers.

References

- Alizamir S, de Véricourt F, Wang S (2020) Warning against recurring risks: An information design approach. *Management Sci.* 66(10):4612–4629.
- Anunrojwong J, Iyer K, Manshadi V (2023) Information design for congested social services: Optimal need-based persuasion. *Management Sci.* 69(7):3778–3796.
- Banerjee AV (1992) A simple model of herd behavior. *Quart. J. Econom.* 107(3):797–817.
- Bassett TJ, Walter CM (2001) Booksellers and bestsellers: British book sales as documented by the bookman, 1891–1906. *Book History* 4(1):205–236.
- Bikhchandani S, Hirshleifer D, Welch I (1992) A theory of fads, fashion, custom, and cultural change as informational cascades. *J. Political Econom.* 100(5):992–1026.
- Boleslavsky R, Cotton CS, Gurnani H (2017) Demonstrations and price competition in new product release. *Management Sci.* 63(6):2016–2026.
- Branco F, Sun M, Villas-Boas JM (2016) Too much information? Information provision and search costs. *Marketing Sci.* 35(4):605–618.
- Cai H, Chen Y, Fang H (2009) Observational learning: Evidence from a randomized natural field experiment. *Amer. Econom. Rev.* 99(3):864–882.
- Callander S, Hörner J (2009) The wisdom of the minority. *J. Econom. Theory* 144(4):1421–1439.
- Calvo E, Cui R, Wagner L (2023) Disclosing product availability in online retail. *Manufacturing Service Oper. Management* 25(2):427–447.
- Candogan O, Drakopoulos K (2020) Optimal signaling of content accuracy: Engagement vs. misinformation. *Oper. Res.* 68(2):497–515.
- Carare O (2012) The impact of bestseller rank on demand: Evidence from the app market. *Internat. Econom. Rev.* 53(3):717–742.
- Compiani G, Lewis G, Peng S, Wang P (2024) Online search and optimal product rankings: An empirical framework. *Marketing Sci.* 43(3):615–636.
- Derakhshan M, Golrezaei N, Manshadi V, Mirrokni V (2022) Product ranking on online platforms. *Management Sci.* 68(6):4024–4041.
- Drakopoulos K, Jain S, Randhawa R (2020) Persuading customers to buy early: The value of personalized information provisioning. *Management Sci.* 67(2):828–853.
- Fu F (2020) The role of sales information in online consumer search. Unpublished PhD thesis, University of Chicago, IL.
- Garcia D, Shelegia S (2018) Consumer search with observational learning. *RAND J. Econom.* 49(1):224–253.
- Glazer J, Kremer I, Perry M (2021) The wisdom of the crowd when acquiring information is costly. *Management Sci.* 67(10):6443–6456.
- Hendricks K, Sorensen A, Wiseman T (2012) Observational learning and demand for search goods. *Amer. Econom. J. Microeconomics* 4(1):1–31.
- Hong H, Shum M (2006) Using price distributions to estimate search costs. *RAND J. Econom.* 37(2):257–275.
- Hu M, Shi M, Wu J (2013) Simultaneous vs. sequential group-buying mechanisms. *Management Sci.* 59(12):2805–2822.
- Hu M, Wang Z, Feng Y (2020) Information disclosure and pricing policies for sales of network goods. *Oper. Res.* 68(4):1162–1177.
- Jin C, Yang L, Hosanagar K (2023) To brush or not to brush: Product rankings, consumer search and fake orders. *Inform. Systems Res.* 34(2):532–552.
- Küçükgül C, Özer Ö, Wang S (2022) Engineering social learning: Information design of time-locked sales campaigns for online platforms. *Management Sci.* 68(7):4899–4918.
- Liu Y, Cooper WL, Wang Z (2019) Information provision and pricing in presence of consumer search costs. *Production Oper. Management* 28(7):1603–1620.
- Moraga-González JL, Sándor Z, Wildenbeest MR (2013) Semi-non-parametric estimation of consumer search costs. *J. Appl. Econom.* 28(7):1205–1223.
- Moraga-González JL, Sándor Z, Wildenbeest MR (2017a) Nonsequential search equilibrium with search cost heterogeneity. *Internat. J. Indust. Organ.* 50:392–414.
- Moraga-González JL, Sándor Z, Wildenbeest MR (2017b) Prices and heterogeneous search costs. *RAND J. Econom.* 48(1):125–146.
- Mueller-Frank M, Pai MM (2016) Social learning with costly search. *Amer. Econom. J. Microeconomics* 8(1):83–109.
- Papanastasiou Y (2020) Fake news propagation and detection: A sequential model. *Management Sci.* 66(5):1826–1846.

- Papanastasiou Y, Savva N (2016) Dynamic pricing in the presence of social learning and strategic consumers. *Management Sci.* 63(4):919–939.
- Sorensen AT (2007) Bestseller lists and product variety. *J. Indust. Econom.* 55(4):715–738.
- Ursu RM (2018) The power of rankings: Quantifying the effect of rankings on online consumer search and purchase decisions. *Marketing Sci.* 37(4):530–552.
- Wang J, Hu M (2020) Efficient inaccuracy: User-generated information sharing in a queue. *Management Sci.* 66(10):4648–4666.
- Wilson H, Macdonald E, Baxendale S (2015) What really makes consumers buy a product. *Harvard Bus. Rev.* (November 9), <https://hbr.org/2015/11/what-really-makes-customers-buy-a-product>.
- Yu M, Debo L, Kapuscinski R (2016) Strategic waiting for consumer-generated quality information: Dynamic pricing of new experience goods. *Management Sci.* 62(2):410–435.

Copyright of Manufacturing & Service Operations Management (M&SOM) (INFORMS) is the property of INFORMS: Institute for Operations Research & the Management Sciences and its content may not be copied or emailed to multiple sites without the copyright holder's express written permission. Additionally, content may not be used with any artificial intelligence tools or machine learning technologies. However, users may print, download, or email articles for individual use.